Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER 2024

Name of the Course: M. Tech (VLSI Design and Systems) Semester: 3rd Semester
Name of the Paper: **Internet of Things** Paper Code: VOM 302

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub-questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	How is the Internet of Things defined, and why do different organizations and frameworks have varying definitions?	**CO1**
(b)	What are the key layers of IoT architecture, and how do they interact to enable IoT functionality?	
(c)	What are the fundamental capabilities required of an IoT node, and how do they contribute to the overall system?	
Q2	(20 marks)	
(a)	What are the primary security and privacy concerns in IoT environments, and what mechanisms are used to mitigate them?	
(b)	What mobility support and communication capabilities are essential for IoT devices, especially in dynamic environments?	**CO2**
(c)	How do RFID and sensor technologies contribute to the functionality and efficiency of IoT systems?	
Q3	(20 marks)	
(a)	What are the basic components of an RFID system, and how does it operate?	**CO3**
(b)	What role do EPCIS and ONS play in the EPCGlobal Architecture Framework for RFID systems?	
(c)	What are the primary security challenges faced by RFID systems, and how can they be mitigated?	
Q4	(20 marks)	
(a)	How do user-centric and device-centric identity management models differ, and what are their respective advantages and challenges?	**CO4**
(b)	What is hybrid identity management, and how does it combine aspects of user-centric and device-centric approaches?	
(c)	Why are identity and trust critical in IoT systems, and what mechanisms are used to ensure both?	
Q5	(20 marks)	
(a)	What are the common vulnerabilities in IoT systems, and how can they be exploited by attackers?	**CO5**
(b)	What are the key security requirements for IoT systems to ensure data protection and system integrity?	**CO5**
(c)	How does the IoT security tomography and layered attacker model help in identifying and mitigating threats?	**CO6**